

SIJIE LIU

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EDUCATION

Carnegie Mellon University

Jan 2026 - Dec 2027

Master of Science in Electrical and Computer Engineering

GPA: 3.67/4.00

Relevant Courses: Generative AI, Machine Learning for Engineers, Computer Vision

Renmin University of China

Sep 2021 - Jun 2025

Bachelor of Engineering in Artificial Intelligence

GPA: 3.70/4.00

Relevant Courses: Machine Learning, Data Mining, Deep Learning, Natural Language Processing, Information Retrieval

PROFESSIONAL EXPERIENCE

Tencent - Interactive Entertainment Group

Shenzhen, China

Machine Learning Engineering Intern

May 2026 - Aug 2026

- Co-developed the **HyRA harness for autocreative**, an self-evolving agent for game ad creative generation running an agent loop that explores, evaluates, and iterates on video-ad concepts over long-horizon workflows. Scaled weekly ideation from ~30 to ~300, per-asset turnaround from 2 weeks to 3 days.
- Explored training-free agent adaptation across creative briefs via scope-aware strategy retrieval from a revision-controlled Experience Registry. Grounded on a versioned trajectory pipeline logging the full workflow and feedback lifecycle: brief interpretation, tool calls, human revisions, campaign outcomes.
- Designed a human-in-the-loop evaluation workflow combining CTR/CPI/ROAS prediction-based proxy scores with pairwise human preferences to rank and refine generated ad hooks, inform scope-aware experience updates, and evaluate experience transfer on held-out creative briefs.

Tencent - Corporate Development Group

Beijing, China

Machine Learning Engineering Intern

Dec 2024 - Jun 2025

- Built a **hierarchical RAG system** for long-form SEC financial statements, combining structure-aware embeddings, Faiss dense retrieval, and TOC-based section routing to address context loss from flat chunking.
- Encoded heading-path prefixes into paragraph embeddings, converted tables into searchable summaries via PaddleOCR + LLM summary, and fused normalized section-routing scores with paragraph-level vector similarities.
- Benchmarked the system using finance-specialized BAM embeddings and jina-embeddings-v3 on FinanceBench, FinQA, and SecQA, achieving consistent gains and improving Recall@10 by 11.6%.

RESEARCH EXPERIENCE

Dense Retrieval for Aggregated Search

Beijing, China

Research Assistant, Advisor: Dr. Jiaxin Mao

Dec 2023 - May 2024

- Built a **prompt-tuned dual-encoder retriever** with contrastive learning for aggregated search across 12.9M WeChat items and 907K high-confidence query-item pairs.
- Designed Graph Transformer structure prompts and global/vertical distribution prompts, improving average Recall@50 by 7.3% over the strongest baseline and 29.7% over DPR across all verticals.
- Publication:** *Dense Retrieval for Aggregated Search* (CIKM 2025), second author.

User Experiment on the Use of ChatGPT

Beijing, China

Research Assistant, Advisor: Dr. Jiaxin Mao

Apr 2023 - Aug 2023

- Led a team of 5 members to conduct a user study on search behavior and learning outcomes under the Search-As-Learning framework, focusing on LLMs as search engines for open-ended tasks.
- Ran a 31-participant experiment across nine task types, combining behavioral logs with GPT-4 and human evaluation to compare traditional, LLM-based, and hybrid search.
- Publication:** *Investigating Users' Search Behavior and Outcome with ChatGPT in Learning-oriented Search Tasks* (SIGIR-AP 2024), first author.

PERSONAL PROJECTS

Anime Character Facial Expression Generation and Replacement

Apr 2026 - May 2026

- Fine-tuned IP-Adapter over a Stable Diffusion 1.5 backbone on ~10k YOLOV8-cleaned image pairs with soft spatial weight masks amplifying facial-keypoint loss to strengthen character identity across expressions.
- Built a best-of-N selection strategy combining ArcFace identity, CLIP semantics, and color-space metrics; lifted expression accuracy from 39.4% to 77.7%.

TECHNICAL SKILLS

Programming: Python, C/C++, SQL, MATLAB

AI & LLM Systems: LLM Agents, MCP, RAG, PyTorch, Hugging Face